FedProp: Federated Graph Neural Networks with Feature Propagation

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Abstract

Graph Neural Networks (GNNs) rely on neighborhood message passing, but this mechanism becomes incomplete in subgraph federated learning when edges connect nodes stored on different clients. These cross-client edges create missing-neighbor information problem: local GNNs cannot directly aggregate features from remote neighbors without additional communication. Existing approaches address this issue by exchanging neighbor information, sharing intermediate embeddings, or learning auxiliary generative models, which can increase communication, computation, or privacy-assumption costs. We introduce FedProp, a model-agnostic preprocessing framework for federated GNNs that treats missing remote-neighbor information as a local feature-imputation problem. Each client constructs an accessible subgraph containing local nodes and boundary placeholders, initializes unknown remote features, and applies boundary-reset feature propagation before standard federated GNN training. FedProp adds no learnable parameters and requires no additional inter-client feature or embedding exchange during training, making it compatible with standard GNN backbones such as GCN and GAT. Theoretically, we ground Laplacian-based propagation in Dirichlet energy minimization. We show that the imputed feature values correspond to the Dirichlet-energy minimizer under fixed local features, and prove linear convergence to this fixed point under stated spectral conditions. Empirically, FedProp substantially improves local-only federated GNN performance across standard node-classification benchmarks and recovers a large fraction of the gap to centralized training while preserving the communication profile of FedAvg. These results show that local feature propagation is a simple and effective mechanism for mitigating cross-client information loss in subgraph federated GNNs.

1 Introduction

Graph Neural Networks (GNNs) are powerful tools for learning from graph-structured data Khemani et al. (2024). By aggregating information from neighboring nodes through message passing Fu et al. (2022), GNNs have achieved strong performance across applications such as social networks, drug discovery, recommender systems, financial graphs, and transportation networks Liu et al. (2022).

In many real-world settings, graph data is decentralized across institutions, devices, or administrative domains, motivating federated learning approaches for graph-structured data Peng et al. (2023); Li et al. (2023; 2025). This paper studies the *subgraph federated learning* setting, where clients collaboratively train a shared GNN over local partitions of a larger graph. Such a setting naturally arises in financial transaction networks, cross-enterprise recommendation systems, and mobility graphs. For example, a bank may know that one of its accounts transacted with an external account, but may not observe the external account's private node features.

The central challenge is that graph partitions are not independent. Edges may connect nodes stored on different clients, creating *cross-client edges*. Since standard GNNs aggregate neighbor features, these edges create a missing-neighbor problem: a client may know that a remote neighbor exists, but cannot directly aggregate that neighbor's features during local training. Ignoring cross-client edges reduces each client to

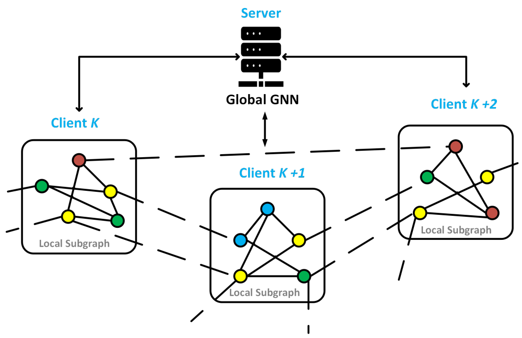


Figure 1: Cross-client edges (dashed lines) connect nodes on distinct clients, leading to missing neighbor information that degrades GNN message passing.

an incomplete induced subgraph, which can substantially degrade node-classification performance relative to centralized training Yao et al. (2023). Figure 1 illustrates this information gap.

Existing federated GNN methods address this problem by either communicating missing neighbor information or learning auxiliary models to generate it. Transmission-based methods improve message-passing fidelity by exchanging neighbor features, embeddings, or related intermediate quantities, while generative methods synthesize missing neighbor information through additional learnable components. These approaches can improve accuracy, but they often introduce extra communication, architecture-specific mechanisms, or additional privacy and trust assumptions.

We introduce **FedProp**, a model-agnostic preprocessing framework for subgraph federated GNNs. FedProp treats missing remote-neighbor information as a local feature-imputation problem. Each client constructs an accessible augmented subgraph containing its local nodes and one-hop remote-neighbor placeholders. The client knows the boundary incidence structure attaching these placeholders to local nodes, but does not observe remote features, remote labels, or remote-remote edges. FedProp initializes the unknown remote-neighbor features and applies boundary-reset feature propagation before standard federated GNN training.

FedProp introduces no additional exchange of raw features, generated features, or intermediate embeddings during training. Its only communication is the standard model-parameter aggregation used by FedAvg McMahan et al. (2023). FedProp is implemented as a local preprocessing step before federated GNN training. Since propagation is performed entirely within each client's accessible subgraph, it does not require exchanging raw node features, generated features, or intermediate embeddings. The resulting completed local features can then be used with standard federated training protocols, including FedAvg, and can be combined with orthogonal privacy mechanisms such as secure aggregation or differential privacy when those guarantees are required.

Theoretically, we groun the algorithm in Dirichlet energy minimization. Under fixed local-feature constraints, the imputed remote-neighbor features correspond to the smoothest graph signal on the accessible augmented subgraph. We further show that the boundary-reset propagation update converges linearly to this fixed point under spectral conditions. This perspective also clarifies FedProp's limitations: when the graph smoothness assumption is weak, as in heterophilic graphs, propagation-based imputation may become unreliable.

Empirically, we evaluate FedProp across standard node-classification benchmarks under IID and non-IID graph partitions. We compare against local-only federated training, oracle-style full-neighborhood variants, and communication-based federated GNN baselines. As summarized in (Figure XYX), FedProp recovers a substantial fraction of the missing-neighbor performance gap while preserving the model-communication profile of FedAvg and avoiding additional feature or embedding exchange.

Our main contributions are:

1. **Feature-communication-free missing-neighbor imputation.** We introduce FedProp, a model-agnostic preprocessing framework that imputes missing remote-neighbor features through boundary-reset feature propagation on an accessible augmented subgraph.
2. **Theory for propagation-based imputation.** We show that Laplacian-based FedProp computes the Dirichlet-energy minimizer under fixed local-feature constraints, prove linear convergence under spectral conditions, and relate reconstruction error to convergence, boundary bias, and irreducible graph-signal mismatch.
3. **Accuracy–communication–regime analysis.** We evaluate FedProp against local-only, oracle, and communication-based baselines, showing when local propagation recovers much of the centralized-performance gap without additional feature exchange and when the smoothness assumption limits performance.

2 Related Work

Federated graph learning. Federated learning enables multiple clients to train a shared model while keeping data decentralized. FedAvg remains the standard communication protocol in many federated systems, where clients train local models and a server aggregates model parameters McMahan et al. (2023). In graph learning, however, federation introduces an additional difficulty: graph structure couples examples across clients. Standard GNNs such as GCN Kipf & Welling (2017), GAT Veličković et al. (2018), and GraphSAGE Hamilton et al. (2018) rely on neighborhood message passing, so partitioning a graph across clients can remove or obscure the neighbors required for local aggregation. Recent surveys and benchmarks distinguish several federated graph learning settings, including graph-level FL, node-level FL, and subgraph federated learning He et al. (2021); **?**. This paper focuses on subgraph-FL, where clients hold local partitions of a larger graph and train a shared GNN under limited cross-client information.

Missing-neighbor handling in subgraph-FL. A central challenge in subgraph-FL is that cross-client edges create missing-neighbor information: a local node may have remote neighbors whose features, labels, or higher-order structure are unavailable to the client. Existing methods address this problem through several mechanisms. Transmission-based methods approximate distributed message passing by communicating information across client boundaries. FedGCN communicates encrypted neighbor feature information in a pre-training step to support federated GCN training Yao et al. (2023), while FedGAT extends this idea to attention-based GNNs through approximations of the quantities needed for graph attention Ambekar et al. (2024). These methods improve message-passing fidelity, but require explicit exchange of cross-client information and are tied to particular GNN backbones.

Generative and inpainting approaches. Another family learns auxiliary models to generate or inpaint missing neighbor information. FedSage+ trains a missing-neighbor generator alongside a federated GraphSAGE model Zhang et al. (2021). FedNI uses network inpainting to recover missing node or edge information in federated population graphs Peng et al. (2023). FedDEP extends this line through deep neighbor generation, embedding prototypes, and an edge-local differential privacy mechanism **?**. These approaches directly target the missing-neighbor problem, but introduce additional learnable components and typically require extra generator training, prototype exchange, or generator-related communication. Moreover, some generative methods use different evaluation protocols, such as Louvain partitioning and inductive splits, which makes direct comparison to Dirichlet-partitioned transductive benchmarks nontrivial.

Coupled, structure-aware, and personalized subgraph-FL. Several recent methods address related problems through coupled-graph modeling, global structural information, or personalized aggregation. FedCog studies federated learning over coupled graphs and exchanges border-node representations between neighboring clients during training to approximate cross-client graph convolution **?**. FedStruct targets interconnected subgraphs using explicit global structural information rather than sharing or generating node features or embeddings **?**. FedGT changes the model class, using a federated graph transformer with global nodes and personalized aggregation to address missing links and subgraph heterogeneity **?**. Other works focus primarily on client heterogeneity rather than missing-feature reconstruction. FED-PUB uses functional embeddings of local GNNs to compute client similarities for personalized aggregation **?**; FedTAD studies topology-aware data-free knowledge distillation under node and topology variation **?**; and AdaFGL, FedGTA, and FedGrAINS address topology heterogeneity, structure non-IIDness, or adaptive neighbor selection through personalized or topology-aware mechanisms **??**Ceyani et al. (2025).

Feature propagation and graph-signal imputation. FedProp is most directly connected to feature propagation for graphs with missing node attributes. Feature propagation exploits graph smoothness: when connected nodes tend to have similar features, missing attributes can be estimated by diffusing known features over the graph. Rossi et al. Rossi et al. (2022) show that this process can be derived from Dirichlet energy minimization and implemented efficiently through iterative diffusion. FedProp adapts this idea to subgraph federated learning, where missing features arise not from random attribute corruption but from graph partitioning across clients. Each client performs boundary-reset propagation on its accessible augmented subgraph, using local features and known boundary incidence to impute missing remote-neighbor features before standard federated GNN training.

Positioning of FedProp. FedProp differs from communication-based, generative, coupled-graph, and personalization-based methods in the role it plays in the learning pipeline. It does not modify the GNN architecture, introduce an auxiliary generator, exchange border embeddings during training, or change the aggregation rule. Instead, FedProp modifies the local input available to a downstream GNN by performing feature propagation as a preprocessing step on each client's accessible augmented subgraph. This makes the method compatible with standard backbones such as GCN and GAT, while avoiding additional exchange of raw features, generated features, or intermediate embeddings beyond FedAvg model aggregation.

3 Preliminaries

Graph notation and GNN message passing. Let $G = (V, E)$ be an undirected graph with adjacency matrix $A \in \mathbb{R}^{|V|\times|V|}$, degree matrix D, and node-feature matrix $X \in \mathbb{R}^{|V|\times d}$. We write $x_v \in \mathbb{R}^d$ for the feature vector of node v, and $N(v)$ for its set of neighbors. Graph Neural Networks (GNNs) learn node representations by iteratively aggregating information from local neighborhoods. A generic message-passing update at layer ℓ can be written as

$$h_v^{(\ell+1)} = f_\theta^{(\ell)}\left(h_v^{(\ell)}, \{h_u^{(\ell)} : u \in N(v)\}, x_v\right), \tag{1}$$

where $h_v^{(\ell)}$ is the representation of node v at layer ℓ, and $f_\theta^{(\ell)}$ is a learnable aggregation function. After L layers, $h_v^{(L)}$ can depend on information from nodes up to L hops away. In matrix form, many GNN layers can be viewed as applying a graph-dependent propagation matrix to node representations, followed by learnable transformations; for example, a GCN layer applies a normalized adjacency operator before a linear map and nonlinearity (Kipf & Welling, 2017).

Graph Laplacians and Dirichlet energy. The unnormalized graph Laplacian is $L = D - A$. The symmetric normalized Laplacian is

$$L_{\text{sym}} = I - D^{-1/2}AD^{-1/2}. \tag{2}$$

For a feature matrix X, the Dirichlet energy with respect to the unnormalized Laplacian is

$$\mathcal{E}(X) = \operatorname{tr}(X^\top L X) = \frac{1}{2}\sum_{(i,j)\in E} A_{ij}\|X_i - X_j\|_2^2. \tag{3}$$

For the symmetric normalized Laplacian, the corresponding normalized energy is

$$\mathcal{E}_{\text{sym}}(X) = \operatorname{tr}(X^\top L_{\text{sym}} X) = \frac{1}{2}\sum_{(i,j)\in E} A_{ij}\left\|\frac{X_i}{\sqrt{D_{ii}}} - \frac{X_j}{\sqrt{D_{jj}}}\right\|_2^2. \tag{4}$$

Low Dirichlet energy indicates that adjacent nodes carry similar feature values. This smoothness view is commonly used in graph signal processing, graph diffusion, and analyses of GNN behavior (Zhou et al., 2021; Rossi et al., 2022).

Feature propagation operators. Feature propagation applies a graph-dependent operator $\widehat{P}$ to a feature matrix over iterative propagation steps:

$$X^{(t+1)} = \widehat{P}X^{(t)}, \tag{5}$$

where t indexes propagation iterations, distinct from the GNN layer index ℓ. Common choices of $\widehat{P}$ include normalized adjacency, heat diffusion, and Personalized PageRank-style propagation (Gasteiger et al., 2022; Rossi et al., 2022). For example, an explicit Euler discretization of heat diffusion with Laplacian L uses

$$\widehat{P}_h = I - hL, \tag{6}$$

where $h > 0$ is the step size. A standard sufficient stability condition for this update is

$$0 < h < \frac{2}{\lambda_{\max}(L)}, \tag{7}$$

where $\lambda_{\max}(L)$ is the largest eigenvalue of L.

Spectral radius and contraction. For a square matrix P, let

$$\rho(P) = \max_i |\lambda_i(P)| \tag{8}$$

denote its spectral radius. If $\rho(P) < 1$, repeated application of P is contractive and $P^t \to 0$ as $t \to \infty$. This spectral condition will be used later to analyze the convergence of boundary-reset feature propagation on the unknown-feature block of each client's accessible graph.

4 Problem Setup and Information Model

Subgraph federated graph. Let $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{X}, \mathbf{Y})$ be an undirected graph with node set $\mathcal{V}$, edge set $\mathcal{E}$, node-feature matrix $\mathbf{X}$, and labels $\mathbf{Y}$ available for a subset of nodes. In subgraph federated learning, the node set is partitioned across K clients,

$$\mathcal{V} = \bigcup_{k=1}^{K} \mathcal{V}_k, \qquad \mathcal{V}_i \cap \mathcal{V}_j = \emptyset \quad \text{for } i \neq j.$$

Client k owns local nodes $\mathcal{V}_k$, their features $\mathbf{X}_{\mathcal{V}_k}$, and labels for a local training subset $\mathcal{T}_k \subseteq \mathcal{V}_k$. The local induced edge set is

$$\mathcal{E}_k^{\text{local}} = \{(u, v) \in \mathcal{E} : u, v \in \mathcal{V}_k\}.$$

Cross-client edges and remote placeholders. Some local nodes may be adjacent to nodes stored by other clients. We define the boundary edge set for client k as

$$\mathcal{E}_k^{\text{bdry}} = \{(v, u) \in \mathcal{E} : v \in \mathcal{V}_k,\ u \notin \mathcal{V}_k\},$$

and the corresponding set of one-hop remote neighbors as

$$\mathcal{U}_k = \{u \notin \mathcal{V}_k : \exists v \in \mathcal{V}_k \text{ such that } (v, u) \in \mathcal{E}\}.$$

Nodes in $\mathcal{U}_k$ are represented locally as remote placeholder nodes. Client k knows the boundary incidence structure connecting $\mathcal{V}_k$ to $\mathcal{U}_k$, but does not observe the remote features $\mathbf{X}_{\mathcal{U}_k}$, remote labels $\mathbf{Y}_{\mathcal{U}_k}$, or remote-remote edges among nodes in $\mathcal{U}_k$. We treat $\mathcal{G}$ as undirected; boundary edges are symmetrized when constructing the local accessible adjacency matrix, so the propagation operator couples local nodes and remote placeholders in both directions.

Accessible augmented graph. Under the primary $L = 1$ information model, client k's accessible augmented graph is

$$\mathcal{G}_k^{\text{acc}} = (\mathcal{V}_k \cup \mathcal{U}_k,\ \mathcal{E}_k^{\text{local}} \cup \mathcal{E}_k^{\text{bdry}}).$$

Table 1: Information available to client k under the strict $L = 1$ and relaxed $L = 2$ topology settings.

Information item	$L = 1$	$L = 2$
Local nodes $\mathcal{V}_k$	Yes	Yes
Local features $\mathbf{X}_{\mathcal{V}_k}$	Yes	Yes
Local training labels $\mathbf{Y}_{\mathcal{T}_k}$	Yes	Yes
Local-local edges $\mathcal{E}_k^{\text{local}}$	Yes	Yes
Boundary incidence $\mathcal{E}_k^{\text{bdry}}$	Yes	Yes
Remote placeholder nodes $\mathcal{U}_k$	Yes	Yes
Remote features $\mathbf{X}_{\mathcal{U}_k}$	No	No
Remote labels $\mathbf{Y}_{\mathcal{U}_k}$	No	No
Remote-remote edges inside $\mathcal{U}_k$	No	Optional topology metadata

Thus, the accessible graph includes both local-local edges and known local-remote boundary edges. It excludes remote-remote edges and all remote features. This distinction is essential: boundary edges are required for local feature propagation to move information from known local nodes into remote placeholder nodes, while excluding remote-remote edges preserves the strict one-hop information model.

Information model. Table 1 summarizes the information available to each client under the strict $L = 1$ setting and the relaxed $L = 2$ topology setting. In this paper, the main experiments use the $L = 1$ model unless otherwise stated. Under $L = 2$, the server may optionally distribute additional topology metadata, such as adjacency indicators among reachable remote nodes; this constitutes a one-time setup communication cost that is separate from the per-round model-parameter communication reported in our efficiency analysis.

Propagation graph, training graph, and loss. FedProp uses the same accessible augmented graph for feature propagation and downstream local GNN training:

$$\mathcal{E}_k^{\text{prop}} = \mathcal{E}_k^{\text{train}} = \mathcal{E}_k^{\text{local}} \cup \mathcal{E}_k^{\text{bdry}}.$$

After imputation, remote placeholders are used only as context nodes for message passing. They do not contribute labels or loss terms. Client k's local training objective is

$$\mathcal{L}_k(\theta) = \sum_{v \in \mathcal{T}_k} \ell\Big(g_\theta(v; \mathcal{G}_k^{\text{acc}}, \widetilde{\mathbf{X}}_k), y_v\Big),$$

where g_θ is a standard message-passing GNN backbone, such as GCN or GAT, and $\widetilde{\mathbf{X}}_k$ denotes the completed feature matrix after propagation. GCN normalization and GAT attention neighborhoods are computed using the edges present in $\mathcal{G}_k^{\text{acc}}$, not the unavailable global graph.

5 The FedProp Method

Overview. FedProp is a local preprocessing step applied independently by each client before federated GNN training. For client k, the input is the accessible augmented graph $\mathcal{G}_k^{\text{acc}}$, the observed local features $\mathbf{X}_{\mathcal{V}_k}$, and the set of remote placeholder nodes $\mathcal{U}_k$. The output is a completed feature matrix $\widetilde{\mathbf{X}}_k$ over $\mathcal{V}_k \cup \mathcal{U}_k$, where local features are kept fixed and remote-placeholder features are imputed through boundary-reset propagation. The completed feature matrix is then used by a standard GNN backbone during local federated training, without requiring any inter-client exchange of features or intermediate representations.

Feature initialization. Let

$$\bar{\mathcal{V}}_k = \mathcal{V}_k \cup \mathcal{U}_k$$

denote the nodes in client k's accessible augmented graph. FedProp constructs an initial feature matrix $\mathbf{X}_k^{(0)} \in \mathbb{R}^{|\bar{\mathcal{V}}_k| \times d}$ as

$$\mathbf{X}_k^{(0)}(v) = \begin{cases} \mathbf{X}_{\mathcal{V}_k}(v), & v \in \mathcal{V}_k, \\ \mathbf{0}, & v \in \mathcal{U}_k. \end{cases} \tag{9}$$

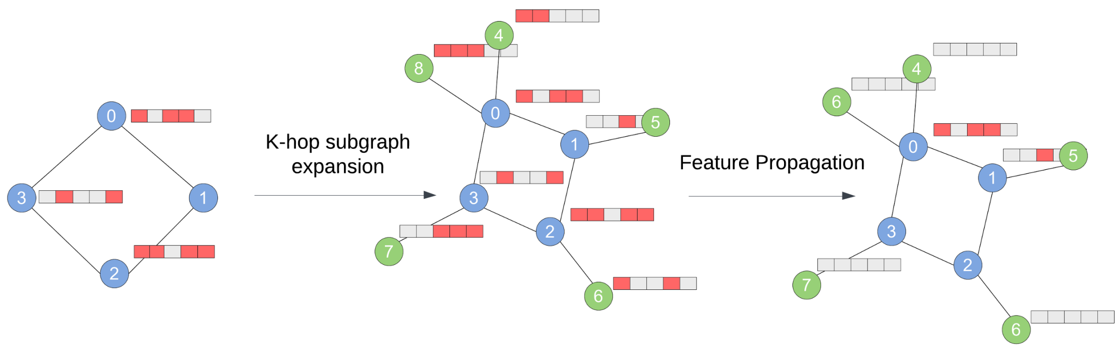


Figure 2: FedProp preprocessing on a client-accessible augmented subgraph. Local node features are observed and fixed, remote-neighbor placeholders are initialized, and feature propagation with boundary reset imputes placeholder features before standard federated GNN training.

Thus, observed local features are retained exactly, while unknown remote-placeholder features are initialized to zero. Other initializations are possible, but zero initialization is used in our experiments unless otherwise stated. We note that the goal of imputation is not exact feature reconstruction, which is information-theoretically impossible without inter-client communication, but rather to produce task-useful feature estimates that restore part of the structural context needed for GNN message passing.

Boundary-reset feature propagation. Let P_k be a propagation operator defined on the accessible augmented graph $\mathcal{G}_k^{\mathrm{acc}}$. FedProp iteratively propagates features and then resets the observed local features:

$$\mathbf{Z}_k^{(t+1)} = P_k \mathbf{X}_k^{(t)}, \tag{10}$$

$$\mathbf{X}_k^{(t+1)}(v) = \begin{cases} \mathbf{X}_{\mathcal{V}_k}(v), & v \in \mathcal{V}_k, \\ \mathbf{Z}_k^{(t+1)}(v), & v \in \mathcal{U}_k. \end{cases} \tag{11}$$

The reset step enforces the boundary condition that observed local features remain fixed throughout propagation. Only the remote-placeholder features are updated.

To make the update explicit, partition P_k according to local nodes $\mathcal{V}_k$ and remote placeholders $\mathcal{U}_k$:

$$P_k = \begin{bmatrix} P_{\mathcal{V}_k\mathcal{V}_k} & P_{\mathcal{V}_k\mathcal{U}_k} \\ P_{\mathcal{U}_k\mathcal{V}_k} & P_{\mathcal{U}_k\mathcal{U}_k} \end{bmatrix}, \qquad \mathbf{X}_k^{(t)} = \begin{bmatrix} \mathbf{X}_{\mathcal{V}_k} \\ \mathbf{X}_{\mathcal{U}_k}^{(t)} \end{bmatrix}. \tag{12}$$

Then the unknown-feature block follows the affine recursion

$$\mathbf{X}_{\mathcal{U}_k}^{(t+1)} = P_{\mathcal{U}_k\mathcal{U}_k} \mathbf{X}_{\mathcal{U}_k}^{(t)} + P_{\mathcal{U}_k\mathcal{V}_k} \mathbf{X}_{\mathcal{V}_k}. \tag{13}$$

When $\rho(P_{\mathcal{U}_k\mathcal{U}_k}) < 1$, this recursion converges to the unique fixed point

$$\mathbf{X}_{\mathcal{U}_k}^{*} = (I - P_{\mathcal{U}_k\mathcal{U}_k})^{-1} P_{\mathcal{U}_k\mathcal{V}_k} \mathbf{X}_{\mathcal{V}_k}.$$

For Laplacian-derived propagation operators, this fixed point corresponds to the harmonic extension of the local boundary features onto the remote placeholders, equivalently the minimizer of the local Dirichlet energy subject to fixed local features. The full convergence and optimality analysis is given in Section A.

Propagation operators. FedProp is a framework rather than a single operator. In this paper, we use two main propagation choices.

The first is normalized-adjacency propagation. Let $\widetilde{A}_k = A_k^{\mathrm{acc}} + I$ be the self-loop-augmented adjacency matrix of $\mathcal{G}_k^{\mathrm{acc}}$, and let $\widetilde{D}_k$ be its degree matrix. The normalized-adjacency operator is

$$P_k^{\mathrm{Adj}} = \widetilde{D}_k^{-1/2} \widetilde{A}_k \widetilde{D}_k^{-1/2}. \tag{14}$$

Algorithm 1 FedProp Preprocessing with Boundary Reset

Require: Accessible augmented graph $\mathcal{G}_k^{\mathrm{acc}}$, local features $\mathbf{X}_{\mathcal{V}_k}$, remote placeholders $\mathcal{U}_k$, propagation operator P_k, maximum iterations T_{prop}, tolerance ε

Ensure: Completed feature matrix $\widetilde{\mathbf{X}}_k$

Initialize $\mathbf{X}_k^{(0)}(v) \leftarrow \mathbf{X}_{\mathcal{V}_k}(v)$ for $v \in \mathcal{V}_k$
Initialize $\mathbf{X}_k^{(0)}(u) \leftarrow \mathbf{0}$ for $u \in \mathcal{U}_k$
for $t = 0$ to $T_{\mathrm{prop}} - 1$ **do**
  $\mathbf{Z}_k^{(t+1)} \leftarrow P_k \mathbf{X}_k^{(t)}$
  $\mathbf{X}_k^{(t+1)}(v) \leftarrow \mathbf{X}_{\mathcal{V}_k}(v)$ for all $v \in \mathcal{V}_k$
  $\mathbf{X}_k^{(t+1)}(u) \leftarrow \mathbf{Z}_k^{(t+1)}(u)$ for all $u \in \mathcal{U}_k$
  if $\|\mathbf{X}_k^{(t+1)} - \mathbf{X}_k^{(t)}\|_F < \varepsilon$ **then**
    break
  end if
end for
$\widetilde{\mathbf{X}}_k \leftarrow \mathbf{X}_k^{(t+1)}$
return $\widetilde{\mathbf{X}}_k$

Although the full normalized-adjacency operator may have spectral radius equal to one, boundary-reset propagation only requires the unknown-node block $P_{\mathcal{U}_k\mathcal{U}_k}$ to be contractive. Intuitively, the fixed local-feature boundary anchors the dynamics, so unknown placeholder nodes converge when they are connected to the observed boundary through the accessible graph.

The second is Laplacian diffusion. Let

$$L_k^{\mathrm{sym}} = I - \widetilde{D}_k^{-1/2} \widetilde{A}_k \widetilde{D}_k^{-1/2}$$

be the symmetric normalized Laplacian of the accessible augmented graph. An explicit Euler diffusion step uses

$$P_k^{\mathrm{Diff}} = I - hL_k^{\mathrm{sym}}, \tag{15}$$

where $h > 0$ is a step size satisfying $0 < h < 2/\lambda_{\max}(L_k^{\mathrm{sym}})$. This condition makes the Euler diffusion update stable; convergence of the boundary-reset iteration further depends on the unknown-node block $P_{\mathcal{U}_k\mathcal{U}_k}$ being a strict contraction. In our experiments, the diffusion step size is fixed across datasets and reported in Section 6.

Repeated application of P_k^{Diff} approximates heat diffusion on the accessible graph. When diffusion is implemented through repeated Euler steps, t applications of $P_k^{\mathrm{Diff}} = I - hL_k^{\mathrm{sym}}$ approximate heat diffusion with effective time $\tau = th$. When the heat kernel is implemented directly, we use the truncated Taylor approximation

$$\exp(-\tau L_k^{\mathrm{sym}}) \approx \sum_{r=0}^{R} \frac{(-\tau L_k^{\mathrm{sym}})^r}{r!}, \tag{16}$$

where R is the Taylor approximation order. We report which diffusion implementation and hyperparameters are used in the experimental protocol.

Stopping criterion. Propagation is run for at most T_{prop} iterations. It terminates early when the feature change falls below a tolerance:

$$\left\| \mathbf{X}_k^{(t+1)} - \mathbf{X}_k^{(t)} \right\|_F < \varepsilon. \tag{17}$$

The propagated matrix at termination is returned as the completed feature matrix $\widetilde{\mathbf{X}}_k$.

Algorithm. Algorithm 1 summarizes FedProp preprocessing for a single client.

Table 2: Datasets used in the experimental evaluation. Cora, Citeseer, Pubmed, and OGBN-Arxiv are used in the main homophilic evaluation; Texas and Wisconsin are used as heterophilic stress tests.

Dataset	**|V|**	**|E|**	**d**	**C**	**Regime**	**Role**
Cora	2,708	5,429	1,433	7	Homophilic	Main
Citeseer	3,327	4,732	3,703	6	Homophilic	Main
Pubmed	19,717	44,338	500	3	Homophilic	Main
OGBN-Arxiv	169,343	1,166,243	128	40	Homophilic	Large-scale main
Texas	183	295	1,703	5	Heterophilic	Limitation study
Wisconsin	251	466	1,703	5	Heterophilic	Limitation study

6 Experiments and Results

Having proven the convergence of our algorithm, we are now ready to present a detailed experimental evaluation of FedProp in terms of node classification accuracy and communications cost for both IID and non-IID partitions of the underlying graph.

6.1 Experimental Protocol

We evaluate FedProp in the subgraph federated node-classification setting described in Section **??**. Unless otherwise stated, all experiments use the strict $L = 1$ information model: each client observes its local nodes, local features, local-local edges, and boundary incidence to one-hop remote placeholders, but does not observe remote features, remote labels, or remote-remote edges. Both feature propagation and downstream GNN training are performed on the accessible augmented graph G_k^{acc}. Remote placeholders are used only as context nodes for message passing and do not contribute labels or loss terms.

Datasets. Our main homophilic evaluation uses four citation benchmarks: Cora, Citeseer, Pubmed, and OGBN-Arxiv. Cora, Citeseer, and Pubmed provide controlled comparison under standard transductive node-classification protocols, while OGBN-Arxiv tests whether FedProp remains effective at larger graph scale. We additionally evaluate Texas and Wisconsin from the WebKB family as heterophilic stress tests. These datasets are included to quantify the failure mode predicted by the smoothness assumption underlying Dirichlet-energy-based propagation. Extended results on Amazon Computers and Amazon Photos are reported in the appendix.

Federated partitioning. Each graph is partitioned into $K = 10$ client subgraphs using a label-based Dirichlet distribution. We use $\beta = 10{,}000$ as the IID setting and $\beta = 10$ as the primary non-IID setting. Lower values of β induce stronger label skew and generally increase the severity of missing-neighbor information. Additional β values are used only in the partition-severity ablation.

Backbones and training. We evaluate FedProp with standard GNN backbones. The main accuracy results use GCN. For Cora, Citeseer, Pubmed, Texas, and Wisconsin, we use a two-layer GCN. For OGBN-Arxiv, we use a larger GCN configuration with three layers and hidden dimension 256, following common OGBN-Arxiv practice. To test model-agnosticity, we also evaluate FedProp with a GAT backbone on the Planetoid datasets. All federated experiments use synchronous FedAvg aggregation with one local epoch per communication round. Results are reported as mean test accuracy and standard deviation over multiple random seeds.

FedProp variants. The main FedProp variants use local boundary-reset feature propagation before federated GNN training. We evaluate two propagation operators. The normalized-adjacency variant uses

$$P_k^{\mathrm{Adj}} = \widetilde{D}_k^{-1/2} \widetilde{A}_k \widetilde{D}_k^{-1/2},$$

where $\widetilde{A}_k = A_k^{\mathrm{acc}} + I$. The diffusion variant uses

$$P_k^{\mathrm{Diff}} = I - hL_k^{\mathrm{sym}},$$

Table 3: Backbone and training settings used in the main experiments. Dataset-specific optimizer and learning-rate details are reported in the appendix.

Dataset group	**Backbone**	**Layers**	**Hidden dim.**	**Dropout**
Cora/Citeseer/Pubmed	GCN	2	16	0.5
Cora/Citeseer/Pubmed	GAT	2	8×8 heads	0.6
OGBN-Arxiv	GCN	3	256	0.5
Texas/Wisconsin	GCN	2	32	0.5

Table 4: Main methods and information used in the experimental comparison. Positional-encoding variants are reported separately in the appendix.

Method	**Propagation**	**Information used**
Centralized	N/A	Full graph and full features
FedProp-Full	N/A	Local graph + true one-hop remote features
FedProp-Zero	None	Local subgraph only
FedProp-Adj	Normalized adjacency	Local features + boundary topology
FedProp-Diff	Discrete diffusion	Local features + boundary topology
FedGCN / FedGAT	Communication-based	Exchanged cross-client information

where L_k^{sym} is the symmetric normalized Laplacian of the accessible augmented graph and $h = 0.1$. Unless otherwise stated, propagation is run for $T_{\mathrm{prop}} = 50$ iterations with boundary reset after every propagation step. Positional-encoding variants are not included in the main tables because their gains are less consistent and can introduce high variance; we report them separately as an appendix ablation.

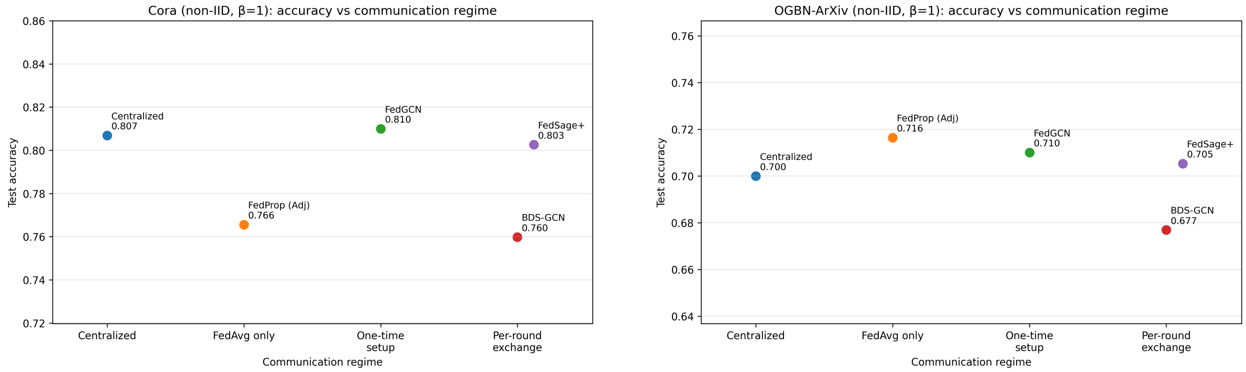
Baselines. We compare FedProp against internal reference systems and communication-based federated GNN baselines. *Centralized* trains the same backbone on the full graph and serves as the centralized reference. *FedProp-Full* is an oracle federated setting in which each client has access to the true one-hop remote features required for message passing. It estimates the best performance attainable if missing features were perfectly available. *FedProp-Zero* performs federated subgraph training without feature imputation and serves as the no-imputation lower reference. We also compare against communication-based methods such as FedGCN and FedGAT where protocol compatibility permits.

Communication accounting. FedProp does not exchange raw features, imputed features, generated features, or intermediate embeddings. Its communication cost is therefore the same as FedAvg: only model parameters are transmitted during aggregation. For communication-based baselines, we separately account for setup or pre-training communication used to exchange neighbor features, polynomial approximations, encrypted quantities, or other cross-client information. This allows us to report both raw accuracy and accuracy–communication tradeoffs.

6.2 Accuracy–Communication Tradeoff

A central claim of FedProp is not merely that it improves accuracy over local subgraph training, but that it does so *without incurring additional inter-client communication beyond standard FedAvg*. We therefore begin the empirical evaluation with the accuracy–communication tradeoff. The relevant comparison is between methods that operate under different communication regimes: purely local methods that preserve the FedAvg communication profile, methods that require a one-time setup exchange of cross-client information, and methods that require repeated communication during training.

Protocol note. The methods compared in this subsection come from different federated graph learning families and are not all evaluated under an identical protocol. In particular, FedProp results use our subgraph federated learning protocol with Dirichlet partitioning, while methods such as FedGCN, BDS-GCN, and FedSage+ are drawn from their reported experimental settings. We therefore interpret the following figures as *communication–accuracy positioning comparisons*, not as strictly controlled apples-to-apples benchmarks.



(a) **Cora** ($\beta = 1$, non-IID).

(b) **OGBN-Arxiv** ($\beta = 1$, non-IID).

Figure 3: Accuracy–communication positioning of representative federated GNN methods. The horizontal axis groups methods by communication regime rather than exact byte count: *FedAvg only*, *one-time setup*, and *per-round exchange*. FedProp (Adj) lies in the FedAvg-only regime because it performs local feature propagation without exchanging raw features, generated features, or intermediate embeddings.

This framing is appropriate for the main purpose of this subsection: to show where FedProp lies in the broader tradeoff landscape.

Communication regimes. For clarity, we group methods into four communication regimes:

- **Centralized**: full-graph training with no federation;
- **FedAvg only**: no additional cross-client exchange beyond standard federated parameter aggregation;
- **One-time setup**: additional pre-training or setup communication is required (e.g., neighbor feature exchange);
- **Per-round exchange**: cross-client information is exchanged repeatedly during training.

FedProp belongs to the *FedAvg only* regime because feature reconstruction is performed locally on each client. By contrast, FedGCN requires a one-time setup communication step, while BDS-GCN and FedSage+ fall into higher-communication regimes due to repeated cross-client exchange during training.

Figure 3 summarizes the tradeoff on a small benchmark (Cora) and a large-scale benchmark (OGBN-Arxiv). These two datasets are sufficient to illustrate the main pattern: FedProp occupies a favorable region of the tradeoff space because it improves substantially over local-only training while avoiding the additional communication required by transmission-based and generative baselines.

On **Cora** (Figure 3a), centralized GCN attains 0.8069 accuracy. FedProp (Adj), which remains in the FedAvg-only regime, reaches 0.7656, outperforming BDS-GCN (0.7598) while avoiding the higher communication burden associated with per-round exchange. FedGCN achieves 0.8100, and FedSage+ reaches 0.8026, but both do so under more communication-intensive settings. Thus, on Cora, FedProp does not achieve the highest raw accuracy, but it offers a strong accuracy–communication tradeoff by remaining entirely communication-free beyond standard FedAvg.

On **OGBN-Arxiv** (Figure 3b), the tradeoff becomes even more compelling. Centralized GCN attains 0.7000 accuracy, while FedProp (Adj) reaches 0.7164 in the FedAvg-only regime. FedGCN achieves 0.7101, BDS-GCN reaches 0.6769, and FedSage+ reaches 0.7053. In this large-scale setting, FedProp not only remains communication-efficient, but also compares favorably in raw accuracy. Although these results should be interpreted with the protocol caveat noted above, they suggest that the communication-free local propagation strategy remains effective beyond the small citation benchmarks.

Table 5: GCN node-classification accuracy under IID Dirichlet partitions. Values are reported as mean accuracy percentage ± standard deviation. FedProp-Adj and FedProp-Diff use local feature propagation without positional encodings.

Method	**Cora**	**Citeseer**	**Pubmed**	**OGBN-Arxiv**
Centralized GCN	80.69 ± 0.65	69.14 ± 0.51	78.50	70.00 ± 0.82
FedProp-Full	80.96 ± 0.63	69.73 ± 0.47	77.94 ± 1.17	70.35 ± 0.81
FedProp-Zero	62.02 ± 1.70	58.78 ± 2.60	73.76 ± 1.15	59.38 ± 0.31
FedProp-Adj	74.77 ± 0.60	65.83 ± 0.55	80.20 ± 0.53	67.81 ± 0.60
FedProp-Diff	75.08 ± 0.81	66.90 ± 0.17	79.80 ± 0.87	67.52 ± 1.80
FedGCN 1-hop	80.09 ± 0.77	69.30 ± 0.69	77.40 ± 0.20	70.04 ± 0.31

Note. FedProp rows report the selected best no-PE accessible-hop setting available for each method and regime. Pubmed centralized GCN is reported without standard deviation because the source summary does not provide one.

Table 6: GCN node-classification accuracy under non-IID Dirichlet partitions. Values are reported as mean accuracy percentage ± standard deviation. Cora, Citeseer, and Pubmed use the primary non-IID setting $\beta = 10$, while OGBN-Arxiv uses the available stronger non-IID setting $\beta = 1$.

Method	**Cora**	**Citeseer**	**Pubmed**	**OGBN-Arxiv**
Centralized GCN	80.69 ± 0.65	69.14 ± 0.51	78.50	70.00 ± 0.82
FedProp-Full	80.19 ± 0.67	69.77 ± 0.79	77.98 ± 0.92	71.14 ± 0.56
FedProp-Zero	61.73 ± 2.14	58.12 ± 1.93	74.47 ± 0.75	64.77 ± 0.15
FedProp-Adj	73.36 ± 1.29	63.44 ± 1.09	77.85 ± 1.38	71.64 ± 0.83
FedProp-Diff	71.29 ± 1.40	62.85 ± 1.43	75.97 ± 0.47	72.14 ± 0.52
FedGCN 1-hop	81.00 ± 0.66	70.06 ± 0.71	77.20 ± 0.30	71.01 ± 0.78

Note. FedProp rows report the selected best no-PE accessible-hop setting available for each method and regime. For Cora, Citeseer, and Pubmed, FedGCN non-IID results are reported from the available $\beta = 1$ source setting because $\beta = 10$ was not reported in the referenced FedGCN results; they should therefore be read as a communication-based reference rather than a perfectly matched controlled baseline.

The main conclusion from this subsection is therefore not that FedProp universally dominates all baselines in raw accuracy. Rather, it occupies an attractive operating point in the design space: it substantially improves over local-only federated training while preserving the communication profile of FedAvg, and it remains competitive with stronger-communication baselines across both small and large graph regimes.

6.3 Main GCN Results: Accuracy and Gap Recovery

Having established the accuracy–communication tradeoff, we now evaluate FedProp under a controlled GCN setting. The purpose of this subsection is to quantify how much of the missing-neighbor performance gap can be recovered by local feature propagation. We compare against three internal references: *Centralized GCN*, which trains on the full graph; *FedProp-Full*, an oracle federated reference with true remote-neighbor features; and *FedProp-Zero*, which performs federated subgraph training without feature imputation. We also include FedGCN 1-hop as a communication-based reference where reported results are available.

Tables 5 and 6 report GCN node-classification accuracy under IID and non-IID partitions, respectively. Across datasets, FedProp-Adj and FedProp-Diff consistently improve over FedProp-Zero, confirming that local feature propagation recovers task-useful information from the accessible boundary structure. The gains are especially meaningful because FedProp introduces no additional inter-client communication beyond the FedAvg communication already required for model aggregation.

In the IID setting, FedProp closes a large fraction of the gap between FedProp-Zero and FedProp-Full. On Cora, FedProp-Diff improves accuracy from 62.02% to 75.08%. On Citeseer, FedProp-Diff improves from 58.78% to 66.90%. On OGBN-Arxiv, FedProp-Adj improves from 59.38% to 67.81%, showing that the approach remains useful at substantially larger graph scale. Pubmed is an interesting case: FedProp-Adj slightly exceeds both the centralized and oracle references. We interpret this conservatively as finite-run variation and possible regularization from propagation, rather than as evidence that the oracle reference is a strict upper bound in every finite experiment.

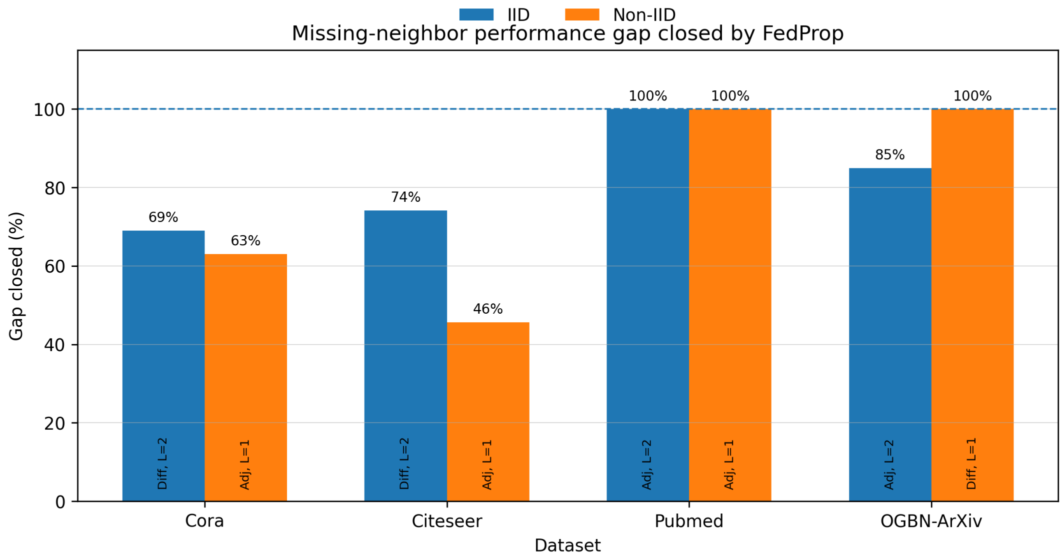


Figure 4: Missing-neighbor performance gap closed by FedProp. For each dataset and partition regime, we report the best no-PE FedProp variant among the available Adj/Diff configurations. Gap closed is computed relative to FedProp-Zero and FedProp-Full. Values are clipped at 100% when the FedProp variant slightly exceeds the oracle reference in finite-run experiments.

The non-IID setting is more challenging because label skew increases the difficulty of local subgraph training and can weaken the boundary information available for propagation. Even in this regime, FedProp remains consistently beneficial. On Cora, FedProp-Adj improves over FedProp-Zero by more than 11 percentage points, increasing from 61.73% to 73.36%. On Citeseer, the gain is smaller but still clear, increasing from 58.12% to 63.44%. On Pubmed, FedProp-Adj nearly matches the oracle reference, reaching 77.85% compared with 77.98% for FedProp-Full. On OGBN-Arxiv, FedProp-Diff reaches 72.14%, outperforming both the FedProp-Full reference and FedGCN 1-hop in the reported summary.

To make these improvements easier to interpret, Figure 4 reports the fraction of the zero-to-oracle performance gap recovered by the best no-PE FedProp variant. We compute

$$\text{GapClosed} = \frac{\text{Acc(FedProp)} - \text{Acc(FedProp-Zero)}}{\text{Acc(FedProp-Full)} - \text{Acc(FedProp-Zero)}} \times 100\%.$$

Values are clipped at 100% for visualization when FedProp slightly exceeds the oracle reference due to finite-run variation.

Figure 4 shows that FedProp recovers a substantial fraction of the missing-neighbor gap across both IID and non-IID regimes. In the IID setting, FedProp closes approximately 69% of the gap on Cora, 74% on Citeseer, 100% on Pubmed, and 85% on OGBN-Arxiv. In the non-IID setting, FedProp closes approximately 63% of the gap on Cora, 46% on Citeseer, 100% on Pubmed, and 100% on OGBN-Arxiv. The lower recovery on Citeseer reflects the harder sparse-graph setting, while the strong Pubmed and OGBN-Arxiv results suggest that local propagation can remain effective when sufficient graph structure is available.

Overall, these results support the central empirical claim of FedProp: local feature propagation recovers a substantial fraction of the accuracy lost by local-only federated subgraph training while preserving the FedAvg communication profile established in Section 6.2. FedGCN remains a strong accuracy baseline, but it relies on explicit cross-client information exchange. FedProp instead targets the low-communication regime, where it consistently improves over FedProp-Zero and often approaches the oracle FedProp-Full reference.

Table 7: GAT node-classification accuracy under IID and non-IID partitions. Values are reported as mean accuracy percentage ± standard deviation. FedProp-GAT variants use the same local propagation procedure as in the GCN experiments, followed by attention-based message passing.

2*Method	**Cora**		**Citeseer**		**Pubmed**	
	IID	**Non-IID**	**IID**	**Non-IID**	**IID**	**Non-IID**
Centralized GAT	83.00	83.00	72.60	72.60	79.00	79.00
FedProp-GAT-Full	81.18 ± 0.81	81.03 ± 0.69	68.30 ± 0.94	69.23 ± 0.87	78.32 ± 0.45	71.19 ± 1.91
FedProp-GAT-Zero	61.30 ± 0.81	67.17 ± 0.76	57.10 ± 1.63	60.04 ± 0.86	64.35 ± 1.20	63.89 ± 2.27
FedProp-GAT-Adj	74.53 ± 2.36	80.18 ± 0.79	65.50 ± 1.78	68.00 ± 1.07	79.36 ± 0.63	71.94 ± 0.42
FedProp-GAT-Diff	70.42 ± 1.11	76.88 ± 1.17	61.78 ± 1.89	63.78 ± 1.88	75.59 ± 1.20	71.32 ± 2.56
DistGAT	64.50 ± 1.20	68.40 ± 1.30	63.50 ± 0.70	66.40 ± 1.40	74.50 ± 0.90	76.90 ± 1.00
FedGAT	80.20 ± 0.30	80.00 ± 0.50	69.40 ± 0.60	69.90 ± 0.40	78.70 ± 0.50	78.90 ± 0.40

Note. Centralized GAT is independent of the federated partition and is therefore repeated across IID and non-IID columns. FedGAT and DistGAT are communication-based GAT references. OGBN-Arxiv is omitted from this table because the current GAT comparison is available only for Cora, Citeseer, and Pubmed; large-scale validation is reported with the GCN backbone in Section 6.3.

6.4 Backbone-Agnosticity: GAT Results

We next evaluate whether FedProp is specific to GCNs or whether the same local propagation mechanism can be paired with a different message-passing architecture. To test this, we replace the GCN backbone with a standard two-layer GAT and repeat the controlled comparison on Cora, Citeseer, and Pubmed. The goal of this experiment is not to claim that FedProp-GAT dominates communication-based GAT methods in raw accuracy, but to test whether feature propagation remains useful when the downstream model uses attention-based aggregation.

Table 7 reports GAT results under IID and non-IID partitions. Across all three datasets, FedProp-Adj and FedProp-Diff improve substantially over FedProp-Zero, showing that the imputed neighbor features remain useful even when the downstream architecture changes from GCN to GAT. This supports the model-agnostic interpretation of FedProp: the propagation step acts as a local preprocessing mechanism for missing-neighbor recovery rather than as a modification tied to a particular GNN layer.

The clearest pattern is the improvement over the zero-imputation GAT baseline. Under IID partitions, FedProp-GAT-Adj improves Cora from 61.30% to 74.53%, Citeseer from 57.10% to 65.50%, and Pubmed from 64.35% to 79.36%. Under non-IID partitions, FedProp-GAT-Adj improves Cora from 67.17% to 80.18%, Citeseer from 60.04% to 68.00%, and Pubmed from 63.89% to 71.94%. These gains indicate that the propagated features are useful to an attention-based GNN, not only to a convolutional one.

The comparison with FedGAT is more nuanced. FedGAT remains a strong communication-based baseline and often achieves higher raw accuracy, especially on Pubmed. However, FedGAT relies on an explicit approximation and communication procedure tailored to attention-based message passing, whereas FedProp-GAT preserves the FedAvg-level communication profile. On Cora non-IID, FedProp-GAT-Adj reaches 80.18%, essentially matching FedGAT at 80.00%. On Citeseer non-IID, FedProp-GAT-Adj reaches 68.00%, approaching FedGAT at 69.90%. Pubmed is the main exception: FedProp-GAT improves substantially over zero-imputation, but remains below FedGAT under non-IID partitions.

Overall, the GAT results support the backbone-agnostic claim in a limited but important sense. FedProp is not a GCN-specific training trick: the same local feature propagation step improves both convolutional and attention-based GNNs. At the same time, the results reinforce the paper's broader framing: FedProp should be understood as a low-communication missing-neighbor recovery method, not as a universal raw-accuracy replacement for communication-based federated GNN algorithms.

6.5 Failure Mode: Heterophilic Graphs

FedProp is motivated by a graph-smoothness assumption: when connected nodes carry compatible features or labels, local propagation can recover task-useful missing-neighbor information. This assumption is natural

Table 8: Heterophilic and weakly homophilic stress-test results under the strict $L = 1$ setting. Values are GCN test accuracy percentages reported as mean $\pm$ standard deviation. Δ_{Zero} reports the improvement of the better FedProp variant over FedProp-Zero. FedProp-Full is shown as an oracle feature-access reference, but it is not treated as a task-performance upper bound under heterophily.

Dataset	Partition	Zero-hop	FedProp-Adj	FedProp-Diff	Δ_{Zero}	Full
Amazon-ratings	IID	37.51 ± 1.79	40.21 ± 0.35	40.06 ± 0.42	$+2.70$	43.27 ± 0.48
Amazon-ratings	Non-IID	35.85 ± 2.40	38.84 ± 0.36	38.66 ± 0.49	$+3.00$	41.16 ± 0.46
Minesweeper	IID	80.00 ± 0.01	80.00 ± 0.00	80.00 ± 0.00	$+0.00$	80.04 ± 0.65
Minesweeper	Non-IID	79.99 ± 0.03	80.00 ± 0.01	80.00 ± 0.01	$+0.01$	80.11 ± 0.52
Roman-empire	IID	55.71 ± 0.33	54.25 ± 0.34	54.93 ± 0.37	-0.77	36.10 ± 0.60
Roman-empire	Non-IID	54.81 ± 0.52	53.22 ± 0.44	53.92 ± 0.35	-0.89	34.48 ± 0.33
Texas	IID	75.83 ± 0.64	72.97 ± 5.02	79.46 ± 2.22	$+3.63$	64.30 ± 3.20
Texas	Non-IID	72.43 ± 1.11	74.46 ± 2.05	74.86 ± 1.27	$+2.43$	62.16 ± 3.51
Wisconsin	IID	70.00 ± 1.12	70.69 ± 1.19	72.75 ± 1.41	$+2.75$	48.73 ± 2.32
Wisconsin	Non-IID	75.88 ± 1.29	74.71 ± 2.00	77.84 ± 1.44	$+1.96$	44.61 ± 2.28

on homophilic citation graphs, but it may fail on heterophilic graphs, where connected nodes often belong to different classes or have dissimilar features. Prior work on heterophilic graph learning has shown that standard message-passing GNNs can be fragile in this regime and may even be outperformed by models that ignore graph structure (**?**). Recent federated graph learning work also emphasizes that homophily heterogeneity can cause local models to learn inconsistent or conflicting structural knowledge across clients (**??**), while broader FGL benchmarking highlights the need to evaluate methods across diverse graph regimes (**?**).

We therefore evaluate FedProp on heterophilic and weakly homophilic datasets to test the limits of the smoothness prior. Unlike the homophilic results in Section 6.3, we do not report zero-to-oracle gap recovery here. In heterophilic graphs, *FedProp-Full* is an oracle feature-access reference, but it is not necessarily an oracle task-performance upper bound. Giving a standard GCN access to true remote-neighbor features can hurt when cross-client edges connect dissimilar nodes, because the GCN aggregation mechanism may mix incompatible signals.

For this reason, we report direct accuracy and the improvement over the no-imputation baseline:

$$\Delta_{\text{Zero}} = \text{Acc}(\text{BestFedProp}) - \text{Acc}(\text{FedProp-Zero}),$$

where BestFedProp is the better of FedProp-Adj and FedProp-Diff for that dataset and partition. Positive values indicate that propagation improves over local-only federated training; negative values indicate that propagation hurts.

Table 8 shows that FedProp behaves differently on heterophilic graphs than on the homophilic citation benchmarks. On Amazon-ratings, FedProp improves over Zero-hop by 2.70 percentage points in the IID setting and 3.00 points in the non-IID setting. In both cases, FedProp moves performance toward the Full reference, but the improvement is modest compared with the large recovery observed on homophilic citation graphs. This suggests that propagation can still recover useful neighbor information under partial heterophily, but the benefit is weaker when the graph signal is less smooth.

Minesweeper shows a different pattern: all methods are essentially tied around 80%. In this case, feature propagation does not materially change the learned decision function. We therefore treat Minesweeper as a neutral result rather than as evidence for or against FedProp.

Roman-empire is the clearest failure case. FedProp underperforms Zero-hop under both IID and non-IID partitions, and FedProp-Full performs substantially worse than Zero-hop. This indicates that additional neighbor information is actively harmful for a vanilla GCN on this graph. In other words, the issue is not only that FedProp fails to reconstruct the Full reference; rather, the Full reference itself is not a desirable target for the downstream GCN. This is precisely the regime predicted by the irreducible-error term in the theoretical decomposition: when the true graph signal contains high-frequency variation across edges, a smooth propagation operator cannot reliably recover task-useful features.

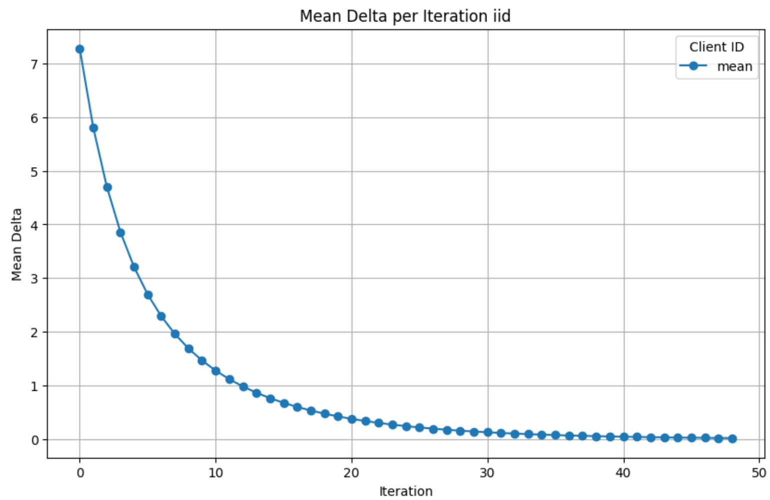


Figure 5: Convergence of the Dirichlet residual $\|\mathcal{L}X\|_F^2$ over propagation iterations under IID partitioning on Cora. The curve reports the mean residual across $K = 10$ clients.

Texas and Wisconsin show a more nuanced intermediate regime. FedProp-Diff improves over Zero-hop on both datasets, with gains of +3.63 and +2.75 points under IID partitions, and +2.43 and +1.96 points under non-IID partitions. However, FedProp-Full is much worse than Zero-hop on both datasets. This confirms that true remote-neighbor feature access is not automatically beneficial under heterophily. FedProp can still provide a useful smoothing/interpolation effect, but the direction of improvement is not well described by the homophilic zero-to-Full gap recovery metric.

Overall, these results qualify the scope of FedProp. FedProp is best understood as a smoothness-based local interpolation method. When the Full reference is beneficial, as in Amazon-ratings, FedProp tends to move performance from Zero-hop toward Full without exchanging features. When Full is harmful, as in Roman-empire and parts of Texas/Wisconsin, FedProp reveals the limitation of smooth propagation rather than acting as a universal improvement mechanism. This supports the broader claim of the paper: FedProp is effective in graph regimes where local smoothness is task-useful, but heterophily-aware propagation or adaptive operator selection is needed when edges systematically connect dissimilar nodes.

6.6 Ablations and Extended Results

We conclude the empirical section with several secondary analyses that support the main results but are not central to the primary accuracy–communication claim. These analyses examine convergence, computational overhead, positional encodings, and extended validation settings.

Convergence and propagation efficiency. To validate the convergence of the local propagation step, we track the Dirichlet residual $\|\mathcal{L}X\|_F^2$ over propagation iterations for each client. Figure 5 reports the mean residual trajectory across $K = 10$ clients on Cora under IID partitioning.

The residual decreases rapidly and stabilizes within approximately 40 iterations across clients. Since feature propagation is performed once as a preprocessing step before federated GNN training, its computational overhead is small relative to the full training procedure. In our experiments, the propagation step accounts for less than 1% of total training time. This supports the practical feasibility of FedProp: the local imputation step improves downstream accuracy while adding only a lightweight sparse-matrix propagation cost.

Propagation depth. All main experiments use $T_{\text{prop}} = 50$ propagation iterations. The convergence behavior in Figure 5 justifies this choice: the residual has largely stabilized before the iteration cap, so additional propagation provides limited benefit while increasing preprocessing cost. Full propagation-depth sensitivity results are reported in Appendix **??**.

Positional encoding ablation. Earlier versions of FedProp included Random Feature Propagation positional encodings. We report these variants in Appendix **??** rather than in the main tables. The reason is empirical: positional encodings sometimes improve accuracy, but the gains are inconsistent and can introduce high variance. To keep the main comparison focused, the primary FedProp results use only the Adj and Diff operators without positional encodings.

Extended datasets. Appendix **??** reports extended validation on Amazon Computers and Amazon Photos. These dense co-purchase graphs differ from the citation benchmarks: local subgraphs already contain substantial neighborhood information, so the gap between FedProp-Zero and the propagation variants is smaller. We therefore treat these results as extended validation rather than as the central empirical story.

$L = 1$ **and** $L = 2$ **accessibility.** The main paper emphasizes the strict $L = 1$ information model, where clients use local nodes, local features, local-local edges, and boundary incidence to one-hop remote placeholders. We also evaluate $L = 2$ variants in Appendix **??**. These can improve performance by exposing richer topology, but they rely on a stronger setup assumption and should not be conflated with the strict zero-additional-communication $L = 1$ setting. For this reason, the main claims are anchored on $L = 1$, while $L = 2$ is treated as an accessibility ablation.

7 Discussion: When FedProp Works and Fails

The results show that FedProp is most effective when the missing-neighbor problem is primarily a missing-feature problem on a graph with useful local smoothness. In the homophilic citation benchmarks, FedProp consistently improves over FedProp-Zero and recovers a substantial fraction of the gap between local-only federated training and the oracle feature-access reference. This supports the central premise of the method: when neighboring nodes tend to carry compatible information, local feature propagation can reconstruct task-useful remote-neighbor features without exchanging raw features, generated features, or intermediate embeddings.

The communication results clarify the operating point of FedProp. Methods such as FedGCN and FedGAT can achieve strong raw accuracy by explicitly communicating additional cross-client information during setup. Generative methods such as FedSage+ address missing neighbors by learning auxiliary generators. FedProp instead occupies the low-communication regime: it keeps the same communication profile as FedAvg and performs all reconstruction locally. The appropriate interpretation is therefore not that FedProp universally dominates all federated GNN methods in raw accuracy, but that it provides a strong accuracy–communication tradeoff when additional cross-client exchange is undesirable or infeasible.

The GAT results further indicate that FedProp is not tied to a specific GNN layer. The same propagation step improves both GCN and GAT backbones over their zero-imputation references. This supports the view of FedProp as a model-agnostic preprocessing mechanism: it modifies the local client input by imputing missing remote-neighbor features, but does not require changing the downstream message-passing architecture.

The heterophilic results qualify this picture. On heterophilic graphs, FedProp-Full is no longer a reliable task-performance upper bound. It is an oracle feature-access reference: it gives the model true remote-neighbor features, but those features can be harmful when the graph connects dissimilar nodes. In this regime, Zero-hop can outperform Full because it avoids smoothing across misleading cross-client edges. The heterophilic experiments therefore show that the missing-neighbor problem and the useful-neighbor problem are not identical. More neighbor information is beneficial only when the downstream GNN can use it productively.

This behavior is consistent with the three-part error decomposition. The transient convergence error decays as propagation proceeds, and the convergence experiments show that this term becomes small within a modest number of iterations. The boundary-bias term depends on the quality and representativeness of the observed local boundary. The irreducible heterophily term remains even after convergence: when the true feature signal varies sharply across edges, no smooth propagation operator can recover the missing features perfectly. Thus, FedProp works best in regimes where local smoothness is task-useful, and its limitations become visible when the graph violates this assumption.

8 Limitations and Future Work

FedProp has several limitations. First, its effectiveness depends on the usefulness of graph smoothness. The method is motivated by Dirichlet-energy minimization and therefore assumes that neighboring nodes contain compatible information. This assumption is reasonable for many homophilic graphs, but it can fail on heterophilic graphs. Our heterophilic experiments show that FedProp can still provide modest gains in some cases, but it can also be neutral or harmful when smoothing mixes incompatible signals. Future work should therefore investigate heterophily-aware propagation operators, adaptive edge weighting, ego-neighbor separation, or learned operators that can decide when propagation should be suppressed.

Second, FedProp reconstructs task-useful features rather than guaranteeing perfect recovery of the true remote features. The goal is not to exactly reproduce hidden neighbor features, which is impossible without additional information, but to produce useful local context for downstream GNN training. A natural extension is to add intrinsic reconstruction diagnostics, such as feature MSE, cosine similarity, spectral fidelity, and per-client recovery statistics. These metrics would separate the quality of feature imputation from the capacity of the downstream GNN.

Third, the method relies on an explicit information model. In the strict $L = 1$ setting, clients observe local nodes, local features, local-local edges, and boundary incidence to one-hop remote placeholders, but not remote features or remote-remote edges. This is the setting that best supports the zero-additional-communication claim. Relaxed $L = 2$ settings can expose richer topology and may improve accuracy, but they require a stronger setup assumption and should be accounted for separately. Future work should study this setup cost more formally and clarify when richer topology is acceptable in practical deployments.

Fourth, FedProp should not be described as providing a formal privacy guarantee by itself. Its core guarantee is narrower and more precise: it introduces no additional inter-client exchange of raw features, generated features, or intermediate embeddings beyond standard federated model aggregation. Formal privacy mechanisms such as secure aggregation or differential privacy could be layered on top of FedProp, but they are not proved in this work.

Fifth, the current evaluation focuses on node classification. This is the standard starting point for subgraph federated GNN evaluation, but it does not cover all graph-learning tasks. Future work should evaluate FedProp on link prediction, graph classification, temporal graphs, recommender systems, and heterogeneous graphs. Another important direction is direct same-codebase evaluation against generative and embedding-sharing baselines such as FedSage+, FedDEP, and FedCog under matched partitioning, splits, and communication accounting.

Finally, while propagation is lightweight in our experiments, local computation may become nontrivial on dense client subgraphs or edge devices with limited resources. The cost scales with the number of propagation iterations, local edges, and feature dimension. Future work should study adaptive stopping criteria, approximate propagation, client-specific iteration budgets, and operator selection under explicit compute constraints.

9 Conclusion

We presented FedProp, a communication-free feature propagation framework for subgraph federated graph learning. FedProp addresses the missing-neighbor problem by treating unavailable remote-neighbor features as a local imputation problem. Each client constructs an accessible augmented graph, initializes unknown remote features, and applies boundary-reset propagation before standard federated GNN training. This procedure introduces no additional inter-client feature exchange and is compatible with standard FedAvg aggregation.

Empirically, FedProp improves substantially over local-only federated training on homophilic citation benchmarks and recovers a large fraction of the missing-neighbor performance gap. The method remains effective at larger scale on OGBN-Arxiv and improves both GCN and GAT backbones, supporting its interpretation as a model-agnostic local preprocessing method. Compared with communication-based federated GNNs, FedProp occupies a different operating point: it trades some raw-accuracy potential for substantially lower communication overhead.

The heterophilic experiments clarify the scope of the method. FedProp is not a universal solution for all graph regimes. When graph edges connect dissimilar nodes, true remote-neighbor features can be harmful to a smoothing GNN, and FedProp-Full should be interpreted as a feature-access reference rather than a strict upper bound. This limitation is not incidental; it follows from the smoothness assumption underlying Dirichlet-energy-based propagation.

Overall, FedProp provides a simple, efficient, and theoretically grounded approach for communication-constrained federated graph learning. It is most appropriate when neighbor information is expected to be task-useful but direct feature exchange is undesirable. Future work should extend the framework with heterophily-aware operators, intrinsic reconstruction diagnostics, stronger privacy accounting, and broader evaluation across federated graph-learning tasks.

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A Analytical Solution and Convergence of Feature Propagation

In this section we show that FedProp's boundary-reset propagation converges to the exact solution of a *Dirichlet feature–imputation* problem on each client's local subgraph. We first derive the closed-form minimizer and then show that the iterative propagator is a strict contraction that reaches that minimizer.

Setting. Client k observes the undirected subgraph $G_k^{(L)} = (V_k \cup U_k, E_k)$, where V_k are *boundary* nodes with known features $X_{V_k} \in \mathbb{R}^{|V_k|\times d}$ and U_k are *interior* nodes whose features $X_{U_k} \in \mathbb{R}^{|U_k|\times d}$ must be imputed. Let L_k be either the combinatorial or normalized Laplacian on $G_k^{(L)}$ and block-partitioned as follows:

$$L_k = \begin{bmatrix} L_{V_kV_k} & L_{V_kU_k} \\ L_{U_kV_k} & L_{U_kU_k} \end{bmatrix}, \qquad X = \begin{bmatrix} X_{V_k} \\ X_{U_k} \end{bmatrix}.$$

Dirichlet SPD condition. Assume every connected component of $G_k^{(L)}$ touches V_k. Then $A := L_{U_kU_k}$ is *symmetric positive definite* (SPD): $A^\top = A$ and $x^\top Ax > 0$ for all $x \neq 0$. Anchoring removes the Laplacian's nullspace, so A is invertible and $0 < \lambda_{\min}(A) \leq \lambda_{\max}(A)$.

A.1 Closed-Form Dirichlet Minimizer

We derive the optimal features X_{U_k} by minimizing the boundary-pinned Dirichlet energy. The objective, with the boundary features X_{V_k} held constant, is to minimize the following residual:

$$\mathcal{R}_k(X_{U_k}) := \tfrac{1}{2}\big\| L_{U_kV_k}X_{V_k} + L_{U_kU_k}X_{U_k} \big\|_F^2.$$

For notational clarity, let $A := L_{U_kU_k}$ and $B := L_{U_kV_k}X_{V_k}$, which allows us to rewrite the objective as $\mathcal{R}_k(X_{U_k}) = \frac{1}{2}\|AX_{U_k} + B\|_F^2$. To find the minimum, we set the gradient with respect to X_{U_k} to zero. Leveraging the symmetry of A, the gradient is:

$$\nabla_{X_{U_k}}\mathcal{R}_k = A^\top(AX_{U_k} + B) = A(AX_{U_k} + B) = 0.$$

Since A is invertible (a consequence of the SPD condition), we can solve for X_{U_k}, yielding the unique minimizer:

$$\boxed{X_{U_k}^\star = -A^{-1}B = -L_{U_kU_k}^{-1}L_{U_kV_k}X_{V_k}}.$$

The uniqueness of this solution is guaranteed by the strict convexity of the objective, confirmed by its Hessian $A^\top A \succ 0$. This closed-form solution is precisely the fixed point to which our iterative propagation converges, as we will demonstrate next.

A.2 Propagation Converges to the Analytical Solution

We now show that FedProp's boundary-reset iteration converges *linearly* to the closed-form minimizer in Eq. equation A.1.

The iteration uses the propagator $P := I - hL_k$ for a chosen step size $h > 0$. At each round t we first update all features, $X^{(t+1)} \leftarrow P\,X^{(t)}$, and then re-clamp the boundary to its ground-truth values, $X_{V_k}^{(t+1)} \leftarrow X_{V_k}^{(0)}$. For the interior nodes this produces the affine recursion

$$X_{U_k}^{(t+1)} = P_{U_kU_k}\,X_{U_k}^{(t)} + P_{U_kV_k}\,X_{V_k}^{(0)}, \tag{18}$$

whose convergence hinges on the linear part $P_{U_kU_k}$.

Contraction of the interior block. Let $A := L_{U_kU_k}$ (SPD), so $P_{U_kU_k} = I - hA$. If λ_i are the eigenvalues of A, those of $P_{U_kU_k}$ are $\mu_i = 1 - h\lambda_i$. Strict contraction, $\|P_{U_kU_k}\|_2 < 1$, requires $|\mu_i| < 1$ for all i, i.e.

$$0 < h < \frac{2}{\lambda_{\max}(A)}.$$

Because $\lambda_{\max}(A) \le \lambda_{\max}(L_k)$, safe choices are easy: for the normalized Laplacian $0 < h < 1$ suffices, while for the combinatorial Laplacian $0 < h < 1/d_{\max}$ is adequate.

Fixed point via Banach. With $P_{U_kU_k}$ a strict contraction, the Banach fixed-point theorem implies that Eq. equation 18 converges linearly to the unique $X^*_{U_k}$ solving

$$(I - P_{U_kU_k})\, X^*_{U_k} = P_{U_kV_k}\, X^{(0)}_{V_k}.$$

Substituting $P_{U_kU_k} = I - hL_{U_kU_k}$ and $P_{U_kV_k} = -hL_{U_kV_k}$ gives

$$\begin{aligned} hL_{U_kU_k}\, X^*_{U_k} &= -\, hL_{U_kV_k}\, X^{(0)}_{V_k} \\ \Longrightarrow &\boxed{X^*_{U_k} = -\, L^{-1}_{U_kU_k} L_{U_kV_k} X^{(0)}_{V_k}}. \end{aligned}$$

Hence Eq. equation A.1 and the fixed point coincide, completing the proof of convergence. In practice the contraction is strong: 30–50 iterations suffice in all experiments.

A.3 Generalisation to Other Symmetric Propagators

The boundary–reset argument is not limited to the linear Laplacian. For any symmetric propagator of the form $P = g(L_k)$ the iteration converges once the interior block is a strict contraction:

$$\rho\big(P_{U_kU_k}\big) < 1 \implies X^{(t)}_{U_k} \longrightarrow X^*_{U_k} = (I - P_{U_kU_k})^{-1} P_{U_kV_k}\, X^{(0)}_{V_k}.$$

Diffusion kernel. For the heat kernel $P_\tau = e^{-\tau L_k}$ the full-graph spectral radius is $\rho(P_\tau) = 1$. Inside FedProp, however, we update only the interior nodes U_k while keeping the boundary nodes V_k fixed. Because the steady-state "all-ones" vector touches the boundary, it cannot live entirely inside U_k. Removing that direction makes the interior block a strict contraction:

$$\rho\big((P_\tau)_{U_kU_k}\big) < 1, \qquad X^{(t)}_{U_k} \longrightarrow X^*_{U_k}.$$

As $\tau \to 0$ this fixed point smoothly approaches the Dirichlet solution $-L^{-1}_{U_kU_k} L_{U_kV_k} X^{(0)}_{V_k}$.

Normalised adjacency. For $\widehat{A} = D^{-1/2}(A_{\text{raw}} + I)D^{-1/2}$ the largest eigenvalue is again 1, but its eigenvector—the degree-weighted constant vector—also touches V_k.

Claim. If every connected component of the unknown-placeholder subgraph is connected to at least one fixed local node through the accessible graph, then the boundary-reset unknown block $\widehat{A}_{U_kU_k}$ is a strict contraction: $\rho(\widehat{A}_{U_kU_k}) < 1$. Indeed, consider any connected component of the accessible graph and let $S \subset V_k \cup U_k$ be its unknown nodes, with the containment proper. Because the component contains at least one fixed local node, S is a proper subset of that component. The corresponding block of $\widehat{A}$ is nonnegative and irreducible with Perron root one, while $\widehat{A}_{SS}$ is a proper principal submatrix that removes at least one nonzero connection to the fixed boundary. By the strict monotonicity of the Perron root for irreducible nonnegative matrices, $\rho(\widehat{A}_{SS}) < 1$. Taking the maximum over unknown components gives $\rho(\widehat{A}_{U_kU_k}) < 1$. Consequently the iteration converges to

$$X^*_{U_k} = (I - \widehat{A}_{U_kU_k})^{-1} \widehat{A}_{U_kV_k} X^{(0)}_{V_k}.$$

Any linear blend $P = (1-h)I + h\widehat{A}$ $(h > 0)$ inherits the same fixed point because $I - P_{U_kU_k} = h(I - \widehat{A}_{U_kU_k})$.

Linear Laplacian (recap). With $P = I - hL_k$ the interior block $P_{U_kU_k} = I - hL_{U_kU_k}$ is a contraction for $0 < h < 2/\lambda_{\max}(L_{U_kU_k})$, and its fixed point is $-L^{-1}_{U_kU_k} L_{U_kV_k} X^{(0)}_{V_k}$, identical to the Dirichlet solution for every admissible step size.

Summary. Anchoring the boundary nodes strips away the non-contracting constant mode from the interior dynamics, turning $P_{U_kU_k}$ into a contraction even when the full propagator has spectral radius 1. The boundary-reset scheme therefore converges linearly for a wide family of propagators, while the linear Laplacian retains the special property of matching the exact Dirichlet minimiser.

B Error Bound for Feature Propagation

This appendix derives a three-term upper bound on the feature reconstruction error. We demonstrate that two seemingly distinct processes—a simplified **global** propagation on the full graph and the practical **client-level** boundary-reset algorithm—can be analyzed through the single, unifying lens of an *affine iterative process*. This unified view reveals that the sources of error are identical in both cases: a transient *propagation error*, a persistent *bias* from the initial conditions, and a fundamental *irreducible error* from graph heterophily.

B.1 The Unified Framework: Affine Iterations

Both propagation dynamics can be described by the general form of an affine iteration:

$$X^{(t+1)} = C\,X^{(t)} + b, \tag{D.1}$$

where C is an operator that must be a strict contraction ($\|C\|_2 < 1$) and b is a constant bias term. By the Banach fixed-point theorem, any such process is guaranteed to converge to the unique fixed point $X^* = (I - C)^{-1}b$.

To analyze the total reconstruction error, $\|X^{(t)} - X_{\text{true}}\|_F$, our strategy is to introduce the fixed point X^* as an intermediate waypoint. The triangle inequality then splits the total error into two components that we can analyze separately:

$$\|X^{(t)} - X_{\text{true}}\|_F \;\le\; \underbrace{\|X^{(t)} - X^*\|_F}_{\text{(1) Propagation Error}} \;+\; \underbrace{\|X^* - X_{\text{true}}\|_F}_{\text{(2) Stationary Error}}. \tag{D.2}$$

B.2 Case 1: Global Propagation Analysis

We first analyze the simplified case of propagation on the full graph, where $X^{(t+1)} = PX^{(t)}$. Here, P is a symmetric propagator with a unique top eigenvalue $\lambda_1 = 1$ and spectral gap $\lambda_{\text{gap}} = \max_{i\ge 2}|\lambda_i| < 1$. The fixed point of this process is the projection onto the stationary subspace: $X^* = P_1X^{(0)}$, where $P_1 = v_1v_1^\top$. To map this to our affine template, we define the operators:

$$C := P - P_1, \quad b := P_1X^{(0)}.$$

(1) Propagation Error. Unrolling the error recursion $E_{t+1} = CE_t$ gives:

$$\|X^{(t)} - X^*\|_F \;\le\; \|C\|_2^t\|X^{(0)} - X^*\|_F \;=\; \lambda_{\text{gap}}^t\,\|(I - P_1)X^{(0)}\|_F. \tag{D.3a}$$

(2) Stationary Error. Introducing P_1X_{true} as an intermediate point gives:

$$\|X^* - X_{\text{true}}\|_F \;\le\; \|P_1(X^{(0)} - X_{\text{true}})\|_F + \|(I - P_1)X_{\text{true}}\|_F. \tag{D.3b}$$

The Complete Global Bound. Combining the propagation and stationary terms via the split in Eq. equation D.2 yields the final three-term bound for the global setting:

$$\boxed{\begin{aligned}\|X^{(t)} - X_{\text{true}}\|_F \le \lambda_{\text{gap}}^t\,&\|(I - P_1)X^{(0)}\|_F\\ &+\|P_1(X^{(0)} - X_{\text{true}})\|_F\\ &+\|(I - P_1)X_{\text{true}}\|_F.\end{aligned}} \tag{D.3c}$$

B.3 Case 2: Client-Level Boundary-Reset Analysis

Now we analyze the practical algorithm for a client k, where the update rule for the interior nodes U_k is already in the affine form $X_{U_k}^{(t+1)} = P_{UU}X_{U_k}^{(t)} + P_{UV}X_{V_k}^{(0)}$.

Setup in the Affine Framework. The operators are defined directly as:

$$C := P_{UU}, \quad b := P_{UV}X_{V_k}^{(0)}, \quad \text{and} \quad X_{U_k}^{*} := (I - P_{UU})^{-1} P_{UV} X_{V_k}^{(0)}.$$

(1) Propagation Error. The rate is governed by $\lambda_{\text{gap}}^{(U)} := \|P_{UU}\|_2 < 1$.

$$\|X_{U_k}^{(t)} - X_{U_k}^{*}\|_F \;\leq\; (\lambda_{\text{gap}}^{(U)})^{\,t}\, \|X_{U_k}^{(0)} - X_{U_k}^{*}\|_F. \tag{D.4a}$$

(2) Stationary Error. We introduce the ideal reconstruction—the harmonic extension of the *true* boundary features:

$$\begin{aligned} \|X_{U_k}^{*} - X_{\text{true},U_k}\|_F \leq{}& \big\|(I - P_{UU})^{-1} P_{UV}\,(X_{V_k}^{(0)} - X_{\text{true},V_k})\big\|_F \\ &+ \big\|(I - P_{UU})^{-1} P_{UV}\, X_{\text{true},V_k} - X_{\text{true},U_k}\big\|_F. \end{aligned} \tag{D.4b}$$

The Complete Client-Level Bound. Combining these terms gives the final, practical error bound for FedProp:

$$\boxed{\begin{aligned} \|X_{U_k}^{(t)} - X_{\text{true},U_k}\|_F \leq{}& \underbrace{(\lambda_{\text{gap}}^{(U)})^{\,t}\, \|X_{U_k}^{(0)} - X_{U_k}^{*}\|_F}_{\text{Propagation Convergence}} \\ &+ \underbrace{\big\|(I - P_{UU})^{-1} P_{UV}\,(X_{V_k}^{(0)} - X_{\text{true},V_k})\big\|_F}_{\text{Initial Boundary Bias}} \\ &+ \underbrace{\big\|(I - P_{UU})^{-1} P_{UV}\, X_{\text{true},V_k} - X_{\text{true},U_k}\big\|_F}_{\text{Irreducible Harmonic Error}}. \end{aligned}} \tag{D.5}$$

B.4 Interpretation and Applicability

The unified analysis reveals that the reconstruction error is always governed by the same three fundamental factors:

- **Propagation Convergence** (Term 1): This transient error decays exponentially. In practice, 30–50 iterations are typically sufficient for it to become negligible.
- **Initial/Boundary Bias** (Term 2): This persistent error arises from a low-frequency mismatch between the initial guess (anchored by the boundary) and the truth. It can be mitigated by better initialization strategies (e.g., mean-centering).
- **Irreducible Heterophily** (Term 3): This fundamental error measures the high-frequency signal in the true features that no smooth, harmonic model can perfectly reconstruct. It is a limit of the method, not a flaw.

This client-level analysis is a strict generalization of the global bound; when the boundary set V_k is empty, Eq. equation D.5 collapses exactly to the global result in Eq. equation D.3c. This theoretical framework applies to any symmetric propagator that satisfies the spectral conditions laid out in Table 9, making the insights broadly applicable across graph learning.

C Extended Experimental Setup and More Results

C.1 Datasets and Statistics

We use the standard homophilic Planetoid citation graphs: Cora, Citeseer, and Pubmed. Table 10 reports node/edge counts, number of classes, feature dimensionality, and average degree.

Operator	**Built from**	**Formula (P)**	**Safe range**
Adj/GCN	$\hat{A}$	$\tilde{D}^{-1/2}\tilde{A}\,\tilde{D}^{-1/2}$	with self-loops
Diffusion	L	$I - hL$	$0 < h < 2/\lambda_{\max}(L)$
APPNP	$\hat{A}$	$(1-\alpha)I + \alpha\hat{A}$	$0 < \alpha < 1$
Heat kernel	L	e^{-tL}	$t > 0$

Table 9: Common symmetric propagators meeting the spectral conditions.

Table 10: **Dataset Statistics.** $|V|$: nodes, $|E|$: edges, C: classes, F: feature dimension, $\bar{d}$: average degree.

Dataset	$\lvert V\rvert$	$\lvert E\rvert$	C	F	$\bar{d}$
Cora	2,708	5,429	7	1,433	4.01
Citeseer	3,327	4,732	6	3,703	2.84
Pubmed	19,717	44,338	3	500	4.50

C.2 Client Partitioning Protocol

We partition each dataset into K client subgraphs using the *label-Dirichlet* scheme from Yao et al. (2023). For class set $\{1, \dots, C\}$, we draw class weights $\boldsymbol{\alpha} \sim \mathrm{Dir}(\beta \mathbf{1}_C)$ *independently per client*, and assign nodes of class c to client k proportionally to $\alpha_c^{(k)}$. We use:

- **IID**: β=10000 (nearly uniform across clients),
- **Non-IID**: β=10 and β=1 (increasing heterogeneity).

This emulates the *missing-neighbor* challenge seen in cross-silo deployments: smaller β yields stronger label imbalance, which increases the fraction of cross-client edges and exacerbates incomplete neighborhoods for local message passing.

C.3 Backbones and Optimization

We couple FedProp with two standard backbones:

GCN (Kipf & Welling, 2017) Two layers, 16 hidden units, ReLU, dropout 0.5 per layer.

GAT (Veličković et al., 2018) Two attention layers, 8 heads $\times$ 8 features (hidden size 64), ELU, dropout 0.6.

Optimization. For Cora/Citeseer we use SGD (lr 0.5, weight decay $5{\times}10^{-4}$); for Pubmed, Adam (lr 0.01, same decay). Training runs up to 600 epochs with early stopping (patience 10) on validation loss. All federated experiments employ FedAvg with synchronous aggregation at each round.

C.4 Feature Propagation Configuration

We reconstruct missing remote-neighbor features locally via iterative propagation on each client's accessible subgraph. We consider two symmetric propagation operators:

- **Adjacency** (*Adj*): $\hat{A} = D^{-\frac{1}{2}}(A{+}I)D^{-\frac{1}{2}}$.
- **Diffusion** (*Diff*): explicit-Euler discretization of the heat equation with the normalized Laplacian $L = I - \hat{A}$, i.e., $X^{(t+1)} \leftarrow (I - hL)\, X^{(t)}$ with Dirichlet boundary reset on known (local) nodes.

Unless stated otherwise, we use a fixed budget of T=50 iterations and step size h=1 (stable for the normalized Laplacian). We terminate early if $\|X^{(t+1)}{-}X^{(t)}\|_F < 10^{-6}$.

C.5 Graph Positional Encoding via Random Feature Propagation

To enrich node features with structural signals, we use *Random Feature Propagation (RFP)* (Eliasof et al., 2023), a lightweight positional encoding method based on diffusing random vectors over the graph.

Let $B^{(0)} \in \mathbb{R}^{n \times R}$ be a matrix of R i.i.d. Gaussian vectors (one per node). We propagate these features over P steps using a symmetric operator P and apply row-wise ℓ_2 normalization at each step:

$$B^{(t+1)} \leftarrow \text{row\_norm}\big(\mathsf{P} \cdot B^{(t)}\big), \quad t = 0, \ldots, P-1,$$

$$B \leftarrow \big[\, B^{(0)} \mid B^{(1)} \mid \cdots \mid B^{(P)} \,\big] \in \mathbb{R}^{n \times R(P+1)}.$$

Defaults. Unless stated otherwise, we set:

$$\mathsf{P} = \hat{A} = D^{-1/2}(A+I)D^{-1/2}, \quad R = 16, \quad P = 5.$$

Usage. We concatenate the resulting positional encoding B to the (reconstructed) node features prior to GNN training.

Cost. Computing RFP incurs $O(P\,|E|\,R)$ time for sparse-dense multiplications and uses $O(n\,R(P+1))$ memory.

C.6 Baselines and Variants

We evaluate:

Centralized GNN trained on the full graph with complete features and edges (upper bound).

FedProp-Zero *No imputation:* remote neighbors are featureless; clients still participate in FedAvg.

FedProp-Full Oracle with exact 1-hop remote features available to each client prior to training.

FedProp (Adj) Feature imputation via $\hat{A}$ for T=50 iterations; no PE.

FedProp (Adj+PE) As above, plus RFP encodings.

FedProp (Diffusion) Feature imputation via $(I - hL)$ iterations (heat equation) with boundary reset; no PE.

FedProp (Diffusion+PE) Diffusion imputation with RFP.

C.7 Evaluation Protocol (Reproducibility)

Unless noted, results are averaged over 10 runs with different seeds; we report mean ± std on the test split. We follow the standard Planetoid splits; hyperparameters are fixed across runs per dataset/setting. *Communication cost* counts only model-parameter exchanges during aggregation; propagation is purely local and incurs no inter-client traffic.

Results with GAT backbone. To further validate the backbone-agnostic design of FedProp, we report node classification results using a Graph Attention Network (GAT). Tables 11 and 12 show that FedProp consistently outperforms the FedProp-Zero baseline and approaches the performance of centralized and full-information models, mirroring the trends observed with GCN backbones.

C.8 Scalability with Respect to Number of Clients

To assess how FedProp scales with data fragmentation, we vary the number of clients from 1 to 20 and evaluate test accuracy on the Cora dataset under both IID and non-IID label partitioning schemes. Figure 6 shows the resulting accuracy trends across different imputation strategies.

Table 11: Node classification accuracy (**mean** $\pm$ **std**) for FedProp-GAT across 10 clients under **IID partition**

Method	Cora	Citeseer	Pubmed
FedProp-Full	0.792 $\pm$ 0.014	0.682 $\pm$ 0.008	0.752 $\pm$ 0.029
FedProp-Zero	0.526 $\pm$ 0.021	0.583 $\pm$ 0.007	0.558 $\pm$ 0.018
FedProp (Adj-PE)	0.729 $\pm$ 0.024	0.648 $\pm$ 0.013	0.756 $\pm$ 0.031
FedProp (Adj)	0.711 $\pm$ 0.022	0.641 $\pm$ 0.015	0.785 $\pm$ 0.022
FedProp (Diffusion-PE)	0.739 $\pm$ 0.021	0.661 $\pm$ 0.008	0.776 $\pm$ 0.018
FedProp (Diffusion)	0.722 $\pm$ 0.016	0.629 $\pm$ 0.023	0.783 $\pm$ 0.020

Table 12: Node classification accuracy (**mean** $\pm$ **std**) for FedProp-GAT across 10 clients under **non-IID partition**

Method	Cora	Citeseer	Pubmed
FedProp-Full	0.784 $\pm$ 0.031	0.686 $\pm$ 0.012	0.767 $\pm$ 0.030
FedProp-Zero	0.506 $\pm$ 0.015	0.599 $\pm$ 0.011	0.595 $\pm$ 0.052
FedProp (Adj-PE)	0.695 $\pm$ 0.019	0.675 $\pm$ 0.008	0.748 $\pm$ 0.019
FedProp (Adj)	0.695 $\pm$ 0.014	0.670 $\pm$ 0.015	0.781 $\pm$ 0.021
FedProp (Diffusion-PE)	0.720 $\pm$ 0.018	0.675 $\pm$ 0.010	0.752 $\pm$ 0.041
FedProp (Diffusion)	0.709 $\pm$ 0.011	0.671 $\pm$ 0.010	0.772 $\pm$ 0.018

As expected, increasing the number of clients leads to a general decline in performance for all methods due to reduced local neighborhood visibility and increased fragmentation. However, this decline is particularly steep for the FedProp-Zero baseline, especially in the non-IID regime, where local label bias exacerbates information loss.

In contrast, FedProp with diffusion-based imputation is highly robust: it preserves high accuracy across a wide range of client counts and shows only a modest decline relative to the centralized upper bound. These results demonstrate that local propagation effectively mitigates the challenges posed by extreme data partitioning in federated graph learning.

C.9 Impact of Positional Encoding (PE)

Positional encodings. While FedProp relies on local feature propagation to reconstruct missing neighbor information, structural biases can further improve learning in fragmented graphs. We explore whether augmenting node features with positional encodings (PE), derived from random feature propagation (RFP), improves performance under different partition regimes and propagation operators.

Role of positional encoding. In addition to reconstructing missing neighbor features via propagation, we investigate the impact of incorporating structural positional encodings. Specifically, we evaluate whether augmenting node features with random feature propagation (RFP) improves classification accuracy across various propagation settings and partition regimes.

Accuracy. Positional encoding (PE) has negligible effect on the zero-hop baseline across both Citeseer and Cora (typically $< 0.3\%$ improvement), as it lacks neighborhood context for propagation. In contrast, both **adjacency-based** and **diffusion-based** propagation variants benefit significantly from PE—gaining up to $+1.8\%$ in accuracy under non-IID settings.

Partition sensitivity. PE has a stronger effect in the non-IID regime ($\beta = 1$), where local structure is highly skewed. Under IID partitions ($\beta = 10000$), clients already receive balanced samples, and the additional inductive signal from PE yields smaller gains.

Recommendation. Given its low computational cost and consistent accuracy improvements in the non-IID setting, we enable PE by default in all remaining experiments unless otherwise stated.

Table 13: Privacy and overhead across federated GNN methods (per round).

Method	**Shared information**	**Structural disclosure risk**	**Client overhead**
FedProp (ours)	Model parameters only	**None** (structure kept local)	Low
FedAvg[†]	Model parameters only	N/A	Low (baseline)
FedGCN	Model params + *neighbor features*	Yes (direct feature exchange)	Moderate
FedGAT	Model params + surrogate metadata	Yes (metadata can leak structure)	Moderate
FedSage+	Model params + *synthetic neighbors*	Yes (synthetics encode structure)	High
FedNI	Model params + GAN-based features	Yes (GAN outputs encode structure)	High

[†] Non-graph baseline (for reference).

C.10 Privacy Analysis and comparison

Deployment assumption (one hop). Each client is assumed to have local access to its one-hop neighborhood (typical in single-provider settings such as a bank ledger, a social platform's first-party graph, or enterprise IT logs). In cross-organization settings, access to cross-entity links requires prior data-sharing agreements and governance.

Privacy summary. FedProp is privacy-by-design: raw features, labels, and edges never leave the client; only model parameters/updates are transmitted, matching the communication surface of standard FL. We assume an honest-but-curious server over secure channels; while inference on updates is possible in principle, FedProp is fully compatible with Secure Aggregation and Differential Privacy to obtain formal guarantees.

Compliance and governance. This design follows data-minimization principles and reduces data-movement and re-identification risk, aligning with regulatory frameworks (e.g., GDPR, CCPA) and domain rules (e.g., HIPAA). In many deployments the graph structure is first-party (available to the provider) while node features/labels remain sensitive; FedProp supports such cases without structural exchange, and can be hardened with SA/DP where policy requires formal guarantees.

At-a-glance comparison. Table 13 contrasts what different federated GNN methods share each round, the resulting structural disclosure risk, and per-client overhead.

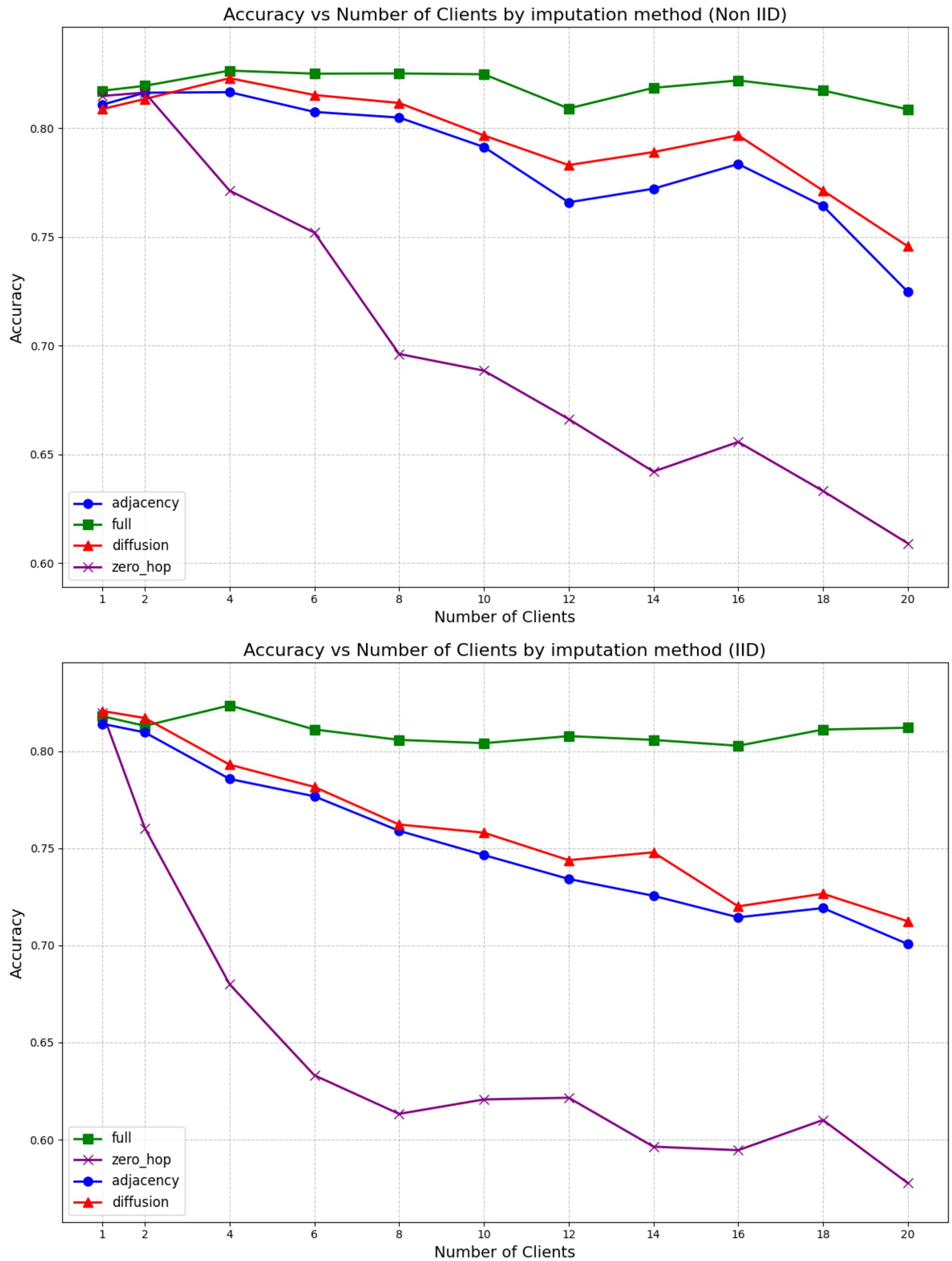
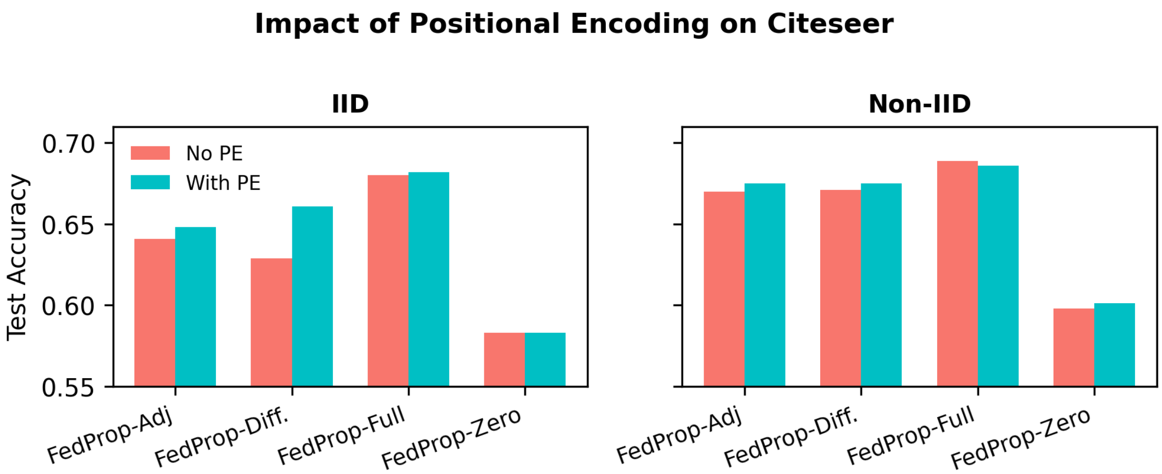
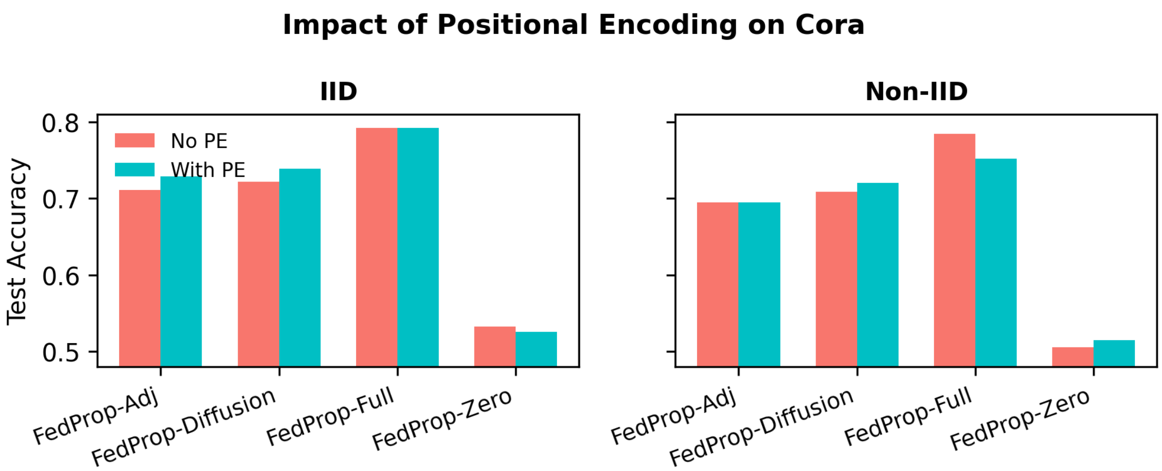


Figure 6: **Scalability with number of clients** on the Cora dataset. Test accuracy under increasing client count for (top) non-IID and (bottom) IID partitions. Zero-hop (no imputation) degrades sharply with fragmentation. In contrast, FedProp with diffusion-based propagation remains stable and closely tracks the oracle (FedProp-Full).



(a) Citeseer



(b) Cora

Figure 7: Impact of positional encoding on client accuracy (averaged over 10 clients) under **non-IID partitioning** ($\beta = 1$). Each bar pair compares test accuracy with and without PE for different propagation operators.